How a Bike Chain Works

01 How can metal bend around a wheel?



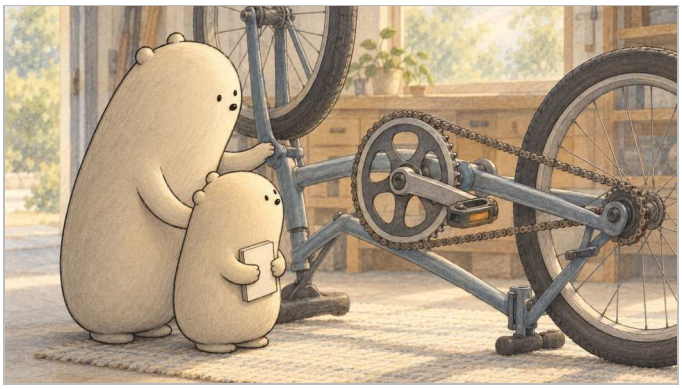
Pins let links bend; teeth catch the chain.

Look closer. Find the little round pins between links.

A little explanation. A chain is a loop of links joined by tiny pins. Each joint lets neighboring links turn. The links stay the same length while the whole loop bends around a sprocket.

Try this. Draw three links joined by dots for pins. Bend your drawn path around a circle.

02 How does the pedal reach the back wheel?



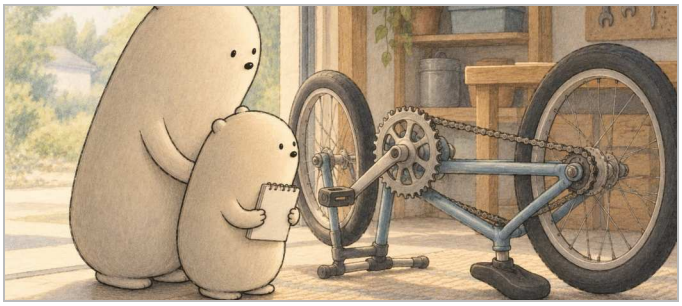
The chain connects the pedal's sprocket to the back wheel's sprocket.

Look closer. Follow the top of the chain between the two sprockets.

A little explanation. A sprocket is a toothed wheel. The front one, called the chainring, turns with the pedals. Its teeth pull the chain; the chain pulls the back sprocket's teeth. That sprocket turns the back wheel as you pedal forward.

Try this. Use the diorama to follow turning, pulling, then turning again.

03 What can a still bike show us?



Count teeth while every part is still.

Look closer. Look for the chain's links and the sprockets' teeth.

A little explanation. Teeth fit into the spaces between the chain's rollers. This fit lets the sprocket pull the chain without slipping. Count the front and back teeth while everything is still.

Try this. Count teeth first, then watch one slow pedal turn. Keep Big beside you. Count only while the bike is still. Big holds an upside-down bike steady on a mat; turn only the pedal slowly by hand, with fingers well away from chain, sprockets, and spokes whenever anything moves. Leave oiling and repairs to Big. At the table, try notched cardboard wheels and a bead loop, or tape paper strips into a chain. Use child scissors only with Big's yes.

04 Can one pedal turn make two wheel turns?



More front teeth can mean more back-wheel turns.

Look closer. Compare the big front sprocket with the small back one.

A little explanation. With 32 teeth in front and 16 behind, one pedal turn gives two back-wheel turns. A big front and small back help with faster riding. A small front and big back make pedaling easier uphill. These choices are gears.

Try this. Draw one circle for a pedal turn and two circles for the back wheel's turns.